Civil Service Analytical Ability: Assumptions, Logic, and Evidence

CIVIL-SERVICE · analytical-logical · BoardPrep lesson v1

Official examination scope

The CSC examination scope covers analytical ability through assumptions and conclusions, logic, word relationships, and data interpretation. This lesson teaches general reasoning methods using original examples; it does not reproduce or predict CSC examination items.

Learning objectives

- Separate a claim from the assumption it needs.
- Test whether a conclusion follows from stated conditions.
- Read comparisons and data without making unsupported causal claims.

1. Claims, assumptions, and evidence

A conclusion is the claim an argument asks you to accept; premises are the reasons offered for it. An assumption is an unstated bridge that must hold for the reasons to support the conclusion. To find it, ask what must be true for the proposed action to work, then test a counterexample. Prefer a necessary, modest assumption over an attractive extra claim. Evidence that two events occurred together supports a description of association, not automatically a causal explanation. Identify alternative causes, changes in measurement, and who entered the comparison before accepting a causal conclusion.

2. Conditional rules and categories

Translate “if P, then Q” as a one-way rule: P is sufficient for Q, and Q is necessary for P. From P you may infer Q; from not-Q you may infer not-P. Q by itself does not prove P, and not-P does not prove not-Q. For category statements, track “all,” “none,” and “some” precisely. “Some” guarantees at least one, not most or all. In a syllogism, test whether the conclusion follows in every case allowed by the premises rather than relying on a familiar real-world association.

3. Relationships, sequences, and transformations

A word analogy is a relation between pairs, not a test of whether two words feel similar. State the relation in a short sentence—such as tool-to-purpose or part-to-whole—then check that it works in the same direction for the second pair. For a number or symbol sequence, test simple differences, ratios, alternating groups, and consistent positional operations. A good rule explains every supplied term with few exceptions. Avoid overfitting one transition or changing rules mid-sequence to force a preferred answer.

4. Data interpretation and comparison

Before comparing figures, identify the unit, denominator, time window, and group being measured. Counts and rates answer different questions; a larger count may simply reflect a larger population. For a change, calculate the difference and choose the stated baseline before calculating a percentage. Check whether a chart’s scale starts at zero or whether categories overlap. State what the data show, what they do not show, and what additional comparison would be needed to support a causal or generalized conclusion.

Find the needed assumption

Question: A city should add a pedestrian crossing near the school because many learners cross there. Which assumption supports the proposal?
1. Identify the proposed action: provide a crossing near the school.
2. Ask what condition makes that location relevant to the action.

3. Check the option against the premise without adding an unrelated safety or cost claim.
Answer: Learners currently cross the road at that location.
Check: If no learners used that location, the stated reason would not support putting a crossing there.

Common misconceptions

- A strong-sounding assumption must be relevant to the conclusion.
- If P implies Q, observing Q proves P.
- A rise after a policy proves the policy caused the rise.

Practice patterns

- State the inference in plain language before evaluating options.
- Test a necessary assumption with a counterexample.
- Name the denominator and comparison group for a data claim.

Official scope references

Official scope: CSC Examination Announcement No. 03 s. 2026: 09 August 2026 CSE-PPT Scope — Philippine Civil Service Commission — https://csc.gov.ph/phocadownload/userupload/erpo/announcements/2026/ExamAnn_2026_03_Conduct%20of%2009%20Aug%202026%20CSE-PPT_orig%20signed.pdf

Instructional references

Instructional reference: Introduction to Logic and Critical Thinking — Matthew J. Van Cleave; Open Textbook Library / University of Minnesota — https://open.umn.edu/opentextbooks/textbooks/introduction-to-logic-and-critical-thinking
Instructional reference: Introductory Statistics 2e — OpenStax; Illowsky and Dean — https://openstax.org/books/introductory-statistics-2e/pages/index

BoardPrep lesson material is original instruction. It is not an official examination document.
No claim about question frequency or examination weighting is made.